

PRODUCT REQUIREMENTS & PRODUCTION READINESS

# AI Operator

## The Media Operating System

**Version** 2.0 · Production Ready · **Date** June 2026  
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**Live** <https://mulbros-marketing-os.onrender.com>  
**Confidential** — prepared for stakeholder review

|                        |                           |                           |                              |
|------------------------|---------------------------|---------------------------|------------------------------|
| <b>34</b><br>AI AGENTS | <b>35</b><br>ACTION TOOLS | <b>14</b><br>INTEGRATIONS | <b>13/13</b><br>PROD LAYERS  |
| <b>28</b><br>DB TABLES | <b>31</b><br>API ROUTES   | <b>107</b><br>COMPONENTS  | <b>47/s</b><br>LOAD BASELINE |

### 1 • Executive Summary

AI Operator is a chat-first, agentic AI operating system that runs the business of talent, agencies, film, and music. Where most "AI assistants" stop at advice, AI Operator **takes real action** — it logs auditions, files royalty splits, builds press kits, and confirms tour dates by calling real APIs through a governed, observable tool layer.

**The one-liner:** One operator, every creative vertical, real actions instead of chat — fully cost-tracked and traceable.

### 2 • Problem & Opportunity

**Problem.** Creative professionals juggle 10+ disconnected SaaS tools. Context is lost between them. Admin busywork eats time that should go to creative work.

**Product.** A single conversational surface where role-specific AI agents take real action across financing, productions, music, touring, and representation — each agent scoped to its domain with its own toolset.

**Moat.** 14 integrations unified behind one governed, observable, rate-limited tool layer — with per-call cost accounting that most agent startups never build. The hard part isn't the chat; it's the safe, audited execution layer underneath.

### 3 • Who It Serves — 3 Personas, 9 Verticals

The interface reshapes itself to the signed-in user. An actor never sees touring tools; a musician never sees casting feeds.

Director / Producer

Film financing, productions, distribution, crew, screenwriting. Agents: lead discovery (live Reddit/Firecrawl), deal analysis, distribution, crew job discovery.

Actor / Talent

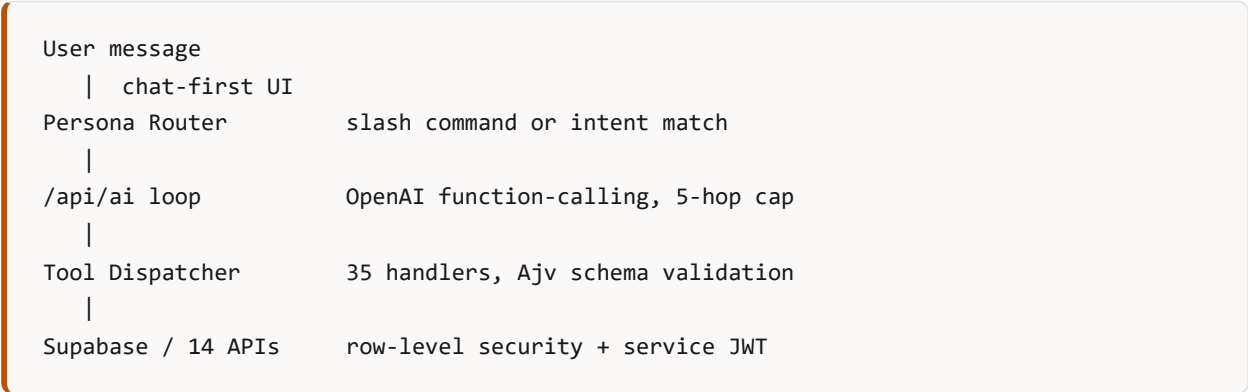
Auditions, self-tape coaching, agent inbox, income & tax, contracts, EPK. Agents: audition tracker, self-tape coach, agent intermediary, income/tax, contract reader.

Musician / Composer

Touring & day-of-show logistics, catalogue + royalty splits, statement audit, sync licensing. Agents: tour manager, catalogue manager, royalty auditor, composer marketing.

Agent distribution (34): Talent 7 · Agency 7 · Cross/Universal 7 · Film & Financing 5 · Music/Composer 4 · Catalogue/Royalty 2 · Touring 1 · EPK 1.

4 • System Architecture



| LAYER         | TECHNOLOGY                                                                                     |
|---------------|------------------------------------------------------------------------------------------------|
| Frontend      | React 18 + Vite · chat-first shell · 107 components · persona-filtered nav · lazy routes       |
| API           | Express 4 · 31 routes · tool-call loop (5-hop cap) · per-tool rate quotas · request-ID tracing |
| Data          | Supabase Postgres · 28 tables · 21 migrations · RLS on every user table                        |
| Auth          | Stytch (email + phone OTP) → Supabase HS256 JWT bridge · email-verified gate · RBAC            |
| Observability | Sentry + Langfuse (auto-priced LLM traces) + cost_ledger — joined on one request ID            |

5 • The Agentic Tool Layer — the differentiator

35 typed, schema-validated, rate-limited tools. Agents don't describe what to do — they call a tool and a row appears in the database.

Tools by domain: Agency 5 · Touring 4 · Catalogue 4 · Payments 4 · Auditions 2 · Comms 2 · EPK 2 · Search/Cost 2.

- 5-hop cap — bounded loop prevents runaway agents
- Ajv schema validation — bad LLM arguments rejected before any handler runs

- **100% cost-tracked** — every call logged to cost\_ledger (provider, model, tokens, USD)
- **HITL approvals** — outbound sends + contract signatures require human approval

6 • Feature Inventory

| MODULE                | CAPABILITY                                                             | STATUS    |
|-----------------------|------------------------------------------------------------------------|-----------|
| Chat-first shell      | Persona router, slash commands, MO avatar, session history             | Live      |
| Agentic tool-calling  | 35 tools, fn-calling loop, HITL approvals, inline tool cards           | Live      |
| Talent suite          | Auditions, self-tape (Mux), agent inbox, income/tax (Plaid), contracts | Live      |
| Agency suite          | Roster, casting feed, submissions, commissions, comms relay            | Live      |
| Touring               | Tours, shows (hold/confirmed), day-of-show logistics, deal terms       | Live      |
| Catalogue & royalties | Releases, tracks, basis-point splits, AI statement parse + anomalies   | Live      |
| EPK builder           | Chat-built press kit, public shareable /epk/:slug page                 | Live      |
| Team chat             | Channels + messages over Supabase Realtime, per-user rate limit        | Live      |
| Film financing        | Live lead discovery (Reddit/Firecrawl), deal analysis agents           | Live      |
| Platform admin        | Cost dashboard, admin-request approval, role escalation                | Live      |
| Cross-session memory  | pgvector agent memory ("MO remembers you")                             | Sprint 13 |
| PDF statement upload  | opendataloader-pdf → statement.parse pipeline                          | Sprint 13 |
| Voice surface         | Talk-to-MO via Pipecat / WebRTC                                        | Sprint 14 |

7 • Delivery Velocity

| SPRINT | DELIVERED                                         |
|--------|---------------------------------------------------|
| 6–7    | Chat-first shell + persona system + Stytch auth   |
| 8      | Agentic tool-calling layer (the foundation)       |
| 9      | Touring module + tour-manager agent               |
| 10     | Catalogue + royalty splits + AI statement parsing |
| 11     | EPK builder + Team chat (Supabase Realtime)       |
| 12     | 13-layer production hardening sweep               |

Cumulative growth: agents 16 → 34 · tools 0 → 35 · tables 12 → 28.

## 8 • Production Readiness — 13 Layers

| #  | LAYER                    | STATUS          |
|----|--------------------------|-----------------|
| 1  | Frontend foundations     | Hardened        |
| 2  | APIs & backend logic     | Hardened        |
| 3  | Database & storage       | Hardened        |
| 4  | Auth & permissions       | Hardened        |
| 5  | Hosting & deployment     | Hardened        |
| 6  | Cloud & compute          | Hardened        |
| 7  | CI/CD & version control  | Hardened        |
| 8  | Security & RLS           | Hardened        |
| 9  | Rate limiting            | Hardened        |
| 10 | Caching & CDN            | Hardened        |
| 11 | Load balancing & scaling | Single-instance |
| 12 | Observability & logs     | Hardened        |
| 13 | Availability & recovery  | Hardened        |

**Security:** Helmet CSP (prod drops unsafe-inline) · RLS everywhere · Styth + 10-min service JWT · Ajv tool-arg validation · tiered + per-tool rate limits · PII scrubbed before observability · CSP violation reporting → Sentry · CORS allowlist · HSTS · X-Frame DENY.

**Reliability:** deep /healthz probe → auto-restart · SIGTERM graceful shutdown · webhook retry-on-failure · 5-min uptime monitor · daily cost-spike alert.

## 9 • Cost & Observability

Unlike most agent products, AI Operator prices every LLM and tool call. Three observability planes join on a single request ID:

```
X-Request-Id (minted per request)
|-- Sentry      errors + session replay
|-- Langfuse    LLM trace, USD per call (auto-priced)
'-- cost_ledger provider, model, tokens, USD
```

- **No surprise bills** — daily threshold alert emails before spend runs away
- **Per-user economics** — know the AI cost-to-serve of every account
- **Per-model breakdown** — route expensive work to cheaper models with data
- **Debuggable** — any slow or failed request traceable end-to-end in seconds

**Pre-launch status:** 9 internal accounts; cost ledger live; Langfuse + Sentry capturing. This document contains zero fabricated usage metrics — all numbers reflect real architecture and capability, not projected traffic.

## 10 • Roadmap

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### Sprint 13 — Near

Cross-session memory (pgvector) · PDF royalty-statement upload · script breakdown tool · PostHog product analytics.

### Sprint 14 — Mid

Voice surface (talk to MO) · self-tape auto-polish (FFmpeg) · MCP server export · OpenTelemetry traces.

### Year 2 — Scale

Multi-tenant architecture · multi-dyno + Redis-backed rate limits · read replicas + CDN tier · microdrama / OTT surface.

## 11 • Summary

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AI Operator is not a demo. It is a **deployed, hardened, observable** platform with a governed agentic action layer — ready for the MulBros team today and architected to scale to multi-tenant tomorrow.

|                    |                                                |
|--------------------|------------------------------------------------|
| Deployed           | Live on Render                                 |
| Layers hardened    | 13 / 13                                        |
| Fabricated metrics | 0 (honest empty states everywhere)             |
| Differentiator     | Governed, cost-tracked, real-action tool layer |

*Confidential · MulBros Media · Prepared by Arghya Chowdhury · June 2026*